

CHEMICAL SYSTEMS, THERMODYNAMICS & MOLECULAR KINETICS

EG-Baccalaureate Advanced STEM Track Handbook

ACADEMIC YEAR: 2025 / 2026 | SENIOR YEAR (EG-BACCALAUREATE ADVANCED DIPLOMA)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced guide to chemical thermodynamics and molecular kinetics: rate laws, catalyst transition state mechanisms, Arrhenius activation energies, reaction quotients Q_c vs K_c , and industrial gas equilibria.

CORE CURRICULUM HIGHLIGHTS:

- Differential and integrated rate laws for zero, first, and second order reactions
- Maxwell-Boltzmann energy distributions and Arrhenius $k = A \exp(-E_a/RT)$
- Perturbation analysis of homogeneous gas phase equilibria under volume and temperature changes
- Industrial Haber-Bosch and Contact process thermodynamic yield optimizations

APPROVED CHAPTER MODULES:

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|---|----------------------------|
| - Reaction Kinetics, Collision Theory & Activation Energy | pp. 1 112 (4 core units) |
| - Dynamic Chemical Equilibria & Le Chatelier Principle | pp. 113 232 (4 core units) |

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Concept & Formula Compendium (8p - 21.6 KB)
Full Ministerial Textbook: 280 Pages - 24.5 MB (Direct on MOE Portal)
Publisher: EG-Baccalaureate Curriculum Advisory Board & MoE Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: EGBAC-CHEM-ADV
Legal Deposit: 2025/14892 - ISBN: 978-977-12-4817-76



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced guide to chemical thermodynamics and molecular kinetics: rate laws, catalyst transition state mechanisms, Arrhenius activation energies, reaction quotients Q_c vs K_c , and industrial gas equilibria. Differential and integrated rate laws for zero, first, and second order reactions Maxwell-Boltzmann energy distributions and Arrhenius $k = A \exp(-E_a/RT)$ Perturbation analysis of homogeneous gas phase equilibria under volume and temperature changes Industrial Haber-Bosch and Contact process thermodynamic yield optimizations

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	40% Understanding & Application Calculations, proofs & models	30% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Reaction Kinetics, Collision Theory & Activation Energy	14 wks	30 pts	50.0%	4 core instructional units covering national standards and
Dynamic Chemical Equilibria & Le Chatelier Principle	14 wks	30 pts	50.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: REACTION KINETICS, COLLISION THEORY & ACTIVATION ENERGY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Reaction Kinetics, Collision Theory & Activation Energy.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Reaction Kinetics, Collision Theory & Activation Energy.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: DYNAMIC CHEMICAL EQUILIBRIA & LE CHATELIER PRINCIPLE

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Dynamic Chemical Equilibria & Le Chatelier Principle.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Dynamic Chemical Equilibria & Le Chatelier Principle.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

CHEMICAL SYSTEMS, THERMODYNAMICS & MOLECULAR KIN

Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$

Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0$, $\sum M_{\text{pivot}} = 0$

Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \text{partial } \rho / \text{partial } t = 0$

Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES

Linear Operator Transformation: $T(\alpha * u + \beta * v) = \alpha * T(u) + \beta * T(v)$

Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$

Sample Variance Boundary: $s^2 = (1 / (n - 1)) * \sum (x_i - \bar{x})^2$

Curricular Efficiency Ratio: $\eta = (\text{Output_actual} / \text{Input_standard})$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION

NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: EGBAC-CHEM-ADV

Curriculum Cycle: 2025/2026 Academic Year | National Edition

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